

LYRA: GENERATIVE 3D SCENE RECONSTRUCTION VIA VIDEO DIFFUSION MODEL SELF-DISTILLATION

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Anonymous submission

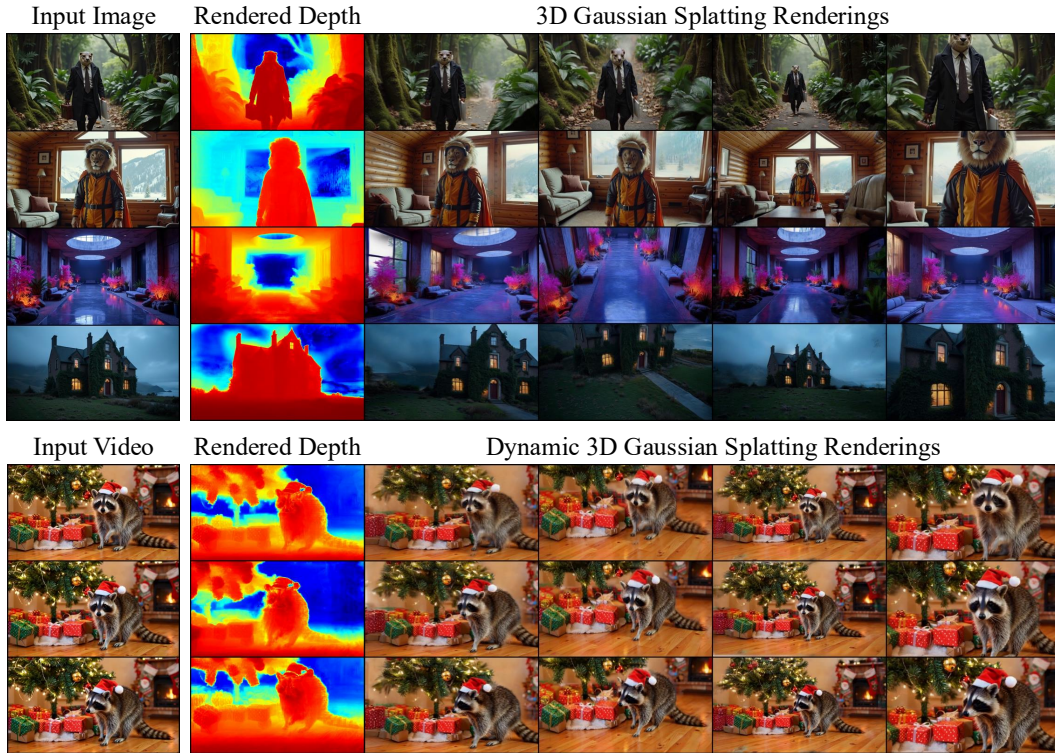


Figure 1: **Feed-Forward 3D and 4D Scene Generation.** From a single image (top), *Lyra* infers a 3D Gaussian Splatting (3DGS) representation in a feed-forward fashion, through self-distilling a video diffusion model without requiring real-world multi-view data. With a video input (bottom), *Lyra* infers a dynamic 3DGS that offers interactive control in both time (rows) and viewpoint (columns).

ABSTRACT

The ability to generate virtual environments is crucial for applications ranging from gaming to physical AI domains such as robotics, autonomous driving, and industrial AI. Current learning-based 3D reconstruction methods rely on the availability of captured real-world multi-view data, which is not always readily available. Recent advancements in video diffusion models have shown remarkable imagination capabilities, yet their 2D nature limits the applications to simulation where a robot needs to navigate and interact with the environment. In this paper, we propose a self-distillation framework that aims to distill the implicit 3D knowledge in the video diffusion models into an explicit 3D Gaussian Splatting (3DGS) representation, eliminating the need for multi-view training data. Specifically, we augment the typical RGB decoder with a 3DGS decoder, which is supervised by the output of the RGB decoder. In this approach, the 3DGS decoder can be purely trained with synthetic data generated by video diffusion models. At inference time, our model can synthesize 3D scenes from either a text prompt or a single image for real-time rendering. Our framework further extends to dynamic 3D scene generation from a monocular input video. Across three standard

benchmarks (RealEstate10K, DL3DV, and Tanks-and-Temples) and every reported metric (PSNR, SSIM, and LPIPS), our framework outperforms all prior single-image-to-3D methods, improving PSNR by up to 3.45 dB over the strongest available baseline on the more challenging out-of-domain benchmarks. On our own diverse evaluation set, it surpasses a strong pixel-space baseline by 8.60 dB PSNR in the static setting and 2.78 dB PSNR in the dynamic setting. These results establish state-of-the-art performance in both static and dynamic 3D scene generation.

1 INTRODUCTION

Creating high-quality 3D environments at scale is a long-standing challenge in computer vision and graphics, empowering applications across film-making, VR/AR, and physical AI with closed-loop simulation. These applications require explicit 3D representations that support real-time rendering, physical interaction, and consistent multi-view synthesis.

Recent advances in neural 3D reconstruction (Mildenhall et al., 2020; Kerbl et al., 2023; Moenne-Loccoz et al., 2024) enable recovering such representations from posed images. However, the reliance on accurate camera poses and high-quality images significantly limits their scalability. The challenge is even greater for dynamic scenes, which typically require synchronized multi-camera setups (Li et al., 2022). Reconstruction methods are also inherently constrained to the observed content and cannot extrapolate beyond the input views. Recent works aim to mitigate these limitations by using feed-forward reconstruction models (Zhang et al., 2024e; Ren et al., 2024c), but these approaches face their own bottleneck: the scarcity of diverse, large-scale 3D training data, which leads to poor out-of-domain generalization. We quantify exactly this gap later: a variant trained only on the standard real multi-view corpora reaches just 19.08 PSNR on diverse out-of-distribution prompts, whereas our diffusion-supervised model reaches 24.77 PSNR on the same evaluation.

Video diffusion models (Cosmos, 2025; Wan, 2025) offer a promising alternative. Trained on massive internet-scale video corpora, they achieve impressive fidelity and generalization across diverse environments. By learning from real-world videos with varied camera trajectories, ranging from handheld motion to cinematic panning and drone footage, these models implicitly encode cues about the underlying 3D world without requiring multi-view training data. Unlike reconstruction approaches (Charatan et al., 2024), generative models can also hallucinate plausible content beyond what is visible in the input frames. Yet, video diffusion models generate only 2D frames, lacking explicit 3D representations. This limits their use in simulation and downstream tasks that demand geometric consistency, long-term coherence, and physical interaction. Encouragingly, recent work (Wang et al., 2024e; Ren et al., 2025) has shown that video models can be adapted for explicit camera control using relatively small-scale posed datasets. This ability to generate posed image sequences transforms them into a powerful tool that can serve as both input and supervision for 3D reconstruction models.

Motivated by this insight, we bridge these two paradigms, reconstruction and generation, through what we call *generative 3D scene reconstruction*. We introduce *Lyra*, a novel method for generating explicit 3D environments from the latent representations of video diffusion models in a single forward pass. Crucially, *Lyra* is trained in a self-distillation framework (Fig. 2), with a 3D Gaussian Splatting (3DGS) decoder operating directly in the latent space of a video diffusion model. Given a single input image or video, we sample a camera trajectory as conditioning input, denoise the resulting video latent, and decode it along two parallel branches. The first branch uses the standard RGB decoder to synthesize a video sequence, while the second is our 3DGS decoder, which produces an explicit 3D representation. Together, these branches form a self-distillation framework in which the RGB branch (teacher) supervises the 3DGS branch (student). To extend viewpoint coverage, we sample multiple camera trajectories, generate multiple video latents, and train the 3DGS decoder to fuse information across them while mitigating long-term and multi-view inconsistencies.

This generative reconstruction framework provides three key benefits, each of which we quantify in our experiments: **(i)** it enables generation of large-scale synthetic environments spanning diverse scenarios directly from video diffusion models, removing the need for real-world multi-view captures—and our ablations show this synthetic supervision alone lifts PSNR from 19.08 to 24.77 over a real-data-only variant; **(ii)** by operating in the latent space of a video model, it scales to $6 \times 121 = 726$ input views at 704×1280 resolution, whereas the equivalent pixel-space decoder simply runs out of memory; and **(iii)** its explicit 3DGS output guarantees geometric consistency, providing representa-

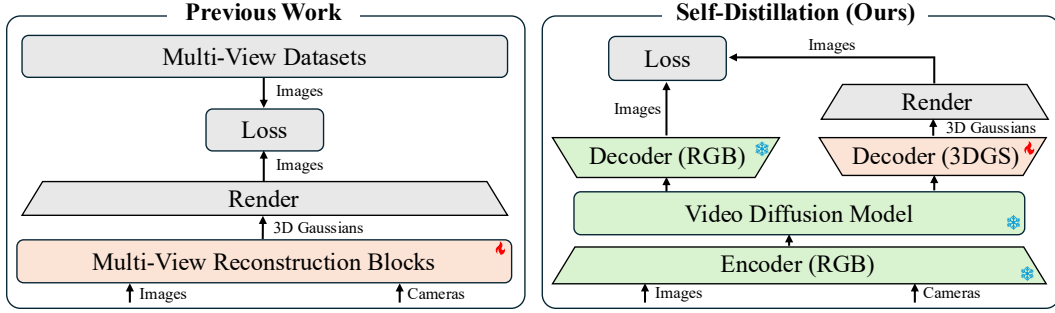


Figure 2: **Self-distillation framework of Lyra.** Previous work (left) (Szymanowicz et al., 2025b) trains multi-view reconstruction blocks using real-world datasets with limited diversity (Zhou et al., 2018; Ling et al., 2024b); our ablations show this real-data-only recipe reaches only 19.08 PSNR on diverse prompts. In contrast, we propose a self-distillation framework (right) for generative 3D scene reconstruction that lifts the same metric to 24.77 PSNR. Precisely, a pre-trained camera-controlled video diffusion model with its RGB decoder output (teacher) supervises the rendering of the 3DGS decoder (student). Using a video model pre-trained on diverse 2D video data allows us to provide diverse multi-view supervision.

tions that are directly applicable to downstream tasks such as physical simulation. Consequently, at inference time, Lyra generates high-quality 3D Gaussian scenes from monocular input that support real-time rendering—decoding a full 704×1280 frame in 18 ms—without requiring any additional optimization or post-processing.

We further extend this self-distillation framework to dynamic 4D generation from monocular video. In this setting, the video model (teacher) provides space-time supervision, while the student learns to produce time-conditioned 3DGS representations that enable novel-view synthesis of dynamic scenes.

Overall, our work makes the following contributions:

- We introduce a self-distillation framework that trains a 3DGS student decoder using a pre-trained camera-controlled video diffusion model as the teacher, eliminating the need for captured multi-view real-world data.
- Our framework extends to dynamic scenes to support 4D reconstruction from a monocular video.
- Our model generalizes across diverse scenes, delivering the best PSNR, SSIM, and LPIPS on all three single-image-to-3D benchmarks simultaneously, and extending the same consistent margins to single-video 4D scene generation—a state-of-the-art result in both settings.

2 RELATED WORK

Multi-view image generation. Early works on multi-view image generation (Watson et al., 2023; Liu et al., 2023; Shi et al., 2024; Wang & Shi, 2023) mainly focus on object-centric scenes without background. Follow-up works (Sargent et al., 2024; Tang et al., 2023b; Schneider et al., 2025) extend multi-view image generators to the scene scale. CAT3D (Gao et al., 2024b) extends a pre-trained image diffusion model for multi-view image generation from any generated or real image input. In a subsequent stage, the generated multi-view images are reconstructed into an explicit 3D representation, e.g., NeRF (Mildenhall et al., 2020) or 3DGS (Kerbl et al., 2023). Moreover, these methods have been extended to generate dynamic 3D scenes from multiple viewpoints using a space-time diffusion model (Watson et al., 2024; Wu et al., 2025b; Kuang et al., 2024; Wang et al., 2025a). Even though these methods demonstrate impressive results, grounding the generations in an explicit 3D representation requires an expensive optimization stage that is not amortized across scenes. Instead of creating multi-view images and then reconstructing using optimization, we are interested in *directly* generating a 3D scene from text or a single image.

Camera-conditioned video generation. There has been significant progress in fine-tuning video diffusion models for 3D camera control. MotionCtrl (Wang et al., 2024e) pioneered camera control by conditioning pre-trained video models with camera poses. Follow-up works (He et al., 2024a; Xu et al., 2024c; Bahmani et al., 2025a;b) represent cameras as Plücker coordinates for pixel-wise conditioning. Another line of work (Hu et al., 2024; Hou et al., 2024; Zhang et al., 2024b) investigates training-free camera control of video diffusion models. Recently, SynCamMaster (Bai et al., 2025b) and ReCamMaster (Bai et al., 2025a) demonstrated impressive camera-controlled video synthesis

using large-scale synthetic training data. While these works achieve compelling results, their output is not grounded in 3D. Our work is orthogonal to this line of work, as we tackle the task of feed-forward 3D generation from camera-controlled video models, allowing novel viewpoint rendering *after* generation. Concretely, we build upon GEN3C (Ren et al., 2025), a recent camera-controlled video diffusion model, and use it as a teacher within a self-distillation framework for 3D reconstruction.

Feed-forward 3D models. Early work on feed-forward 3D reconstruction mainly focused on object-centric scenes without background. These methods directly predict a NeRF (Hong et al., 2024; Li et al., 2024b) or 3DGS (Tang et al., 2024a; Szymanowicz et al., 2024) from either text or an image. More recent works (Charatan et al., 2024; Zhang et al., 2024e; Szymanowicz et al., 2025a; Ren et al., 2024c) focus on scene-level 3D reconstruction, typically regressing per-pixel 3D Gaussians from one or more input images. However, most scene-level 3D reconstruction methods are limited to training distribution scenes, e.g., RealEstate10K (Zhou et al., 2018), and generalize poorly to generated scenes—a gap our ablations make concrete, where a real-data-only variant trails our full model by 5.69 PSNR on diverse prompts. Recent methods extend feed-forward 3D reconstruction models to generate scenes. Bolt3D (Szymanowicz et al., 2025b) trains a pointmap (Wang et al., 2024b) autoencoder to generate pointmaps along the multi-view images, used for feed-forward 3D reconstruction. Closely related to our work, Wonderland (Liang et al., 2025a) uses a camera-controlled video model to generate 3D Gaussians with a feed-forward network; it is the strongest published baseline on all three benchmarks where it reports numbers, yet our method still improves on it by 3.45 PSNR on DL3DV and 3.34 PSNR on Tanks-and-Temples. In contrast to these works, we use a camera-controlled video model as a teacher within a self-distillation framework to train our student 3D model *without requiring real-world multi-view data*. Importantly, we show a simple extension of our framework for feed-forward 4D scene generation, an unexplored task to date.

3 SELF-DISTILLATION USING VIDEO DIFFUSION MODELS

Our key idea is to distill the implicit 3D knowledge embedded in video diffusion models into an explicit 3DGS decoder capable of producing high-quality 3D representations. To this end, we build a teacher–student framework in which the video diffusion model (teacher) generates RGB videos that supervise the 3DGS decoder (student) that operates in the same latent space, as illustrated in Fig. 2. In this section, we detail our video diffusion model backbone (Sec. 3.1) and its role in self-distillation for the 3DGS decoder (Sec. 3.2). We discuss the design choice for the 3DGS decoder in Sec. 4 and outline minimal changes required to adapt the pipeline to dynamic 3D scenes in Sec. 5.

3.1 BACKGROUND: CAMERA-CONTROLLED AND 3D-CONSISTENT VIDEO DIFFUSION

Since we supervise our 3DGS decoder only with the camera-controlled video diffusion model, we heavily rely on its 3D consistency. Due to this, we build our approach on GEN3C (Ren et al., 2025), a recent camera-conditioned video diffusion model. In the following we provide an overview of the key design choices made in GEN3C.

Spatiotemporal 3D cache. To improve video consistency and camera control precision, GEN3C constructs a spatiotemporal 3D cache $\{\mathbf{P}^{t,v}\}$ from input image(s) or videos, where each $\mathbf{P}^{t,v}$ is a colored point cloud obtained by unprojecting the depth estimation (Wang et al., 2024a) of an RGB image captured from camera viewpoint v at time t . The cache is organized as an $L \times V$ array, where L denotes the number of frames (temporal length) and V denotes the number of camera views.

Rendering and structured guidance. To leverage this cache, GEN3C renders each point cloud from arbitrary given camera poses, producing RGB images $\mathbf{I}^{t,v}$ and disocclusion masks $\mathbf{M}^{t,v}$ via $(\mathbf{I}^{t,v}, \mathbf{M}^{t,v}) = \mathcal{R}(\mathbf{P}^{t,v}, \mathbf{C}^t)$, where $\mathcal{R}$ denotes the rendering function that projects the 3D point cloud $\mathbf{P}^{t,v}$ onto the 2D camera plane according to the camera pose $\mathbf{C}^t$ at time t . Given a sequence of camera poses $\mathbf{C} = (\mathbf{C}^1, \dots, \mathbf{C}^L)$, rendering all cache elements produces V videos of length L , which can be stacked into image sequences $\mathbf{I}^v \in \mathbb{R}^{L \times 3 \times H \times W}$ and mask sequences $\mathbf{M}^v \in \mathbb{R}^{L \times 1 \times H \times W}$ for each view v . The disocclusion masks indicate areas that the video diffusion model should fill in. These renderings serve as structured visual guidance for subsequent video generation.

Video variational autoencoder (VAE). In diffusion-based video generation models, a video variational autoencoder (VAE) (Kingma & Welling, 2013; Rombach et al., 2022) is commonly employed to compress videos into a lower-dimensional latent space for efficient training and inference. Given a RGB video $\mathbf{I} \in \mathbb{R}^{L \times 3 \times H \times W}$, a pre-trained VAE encoder $\mathcal{E}$ will encode the video into a latent space,

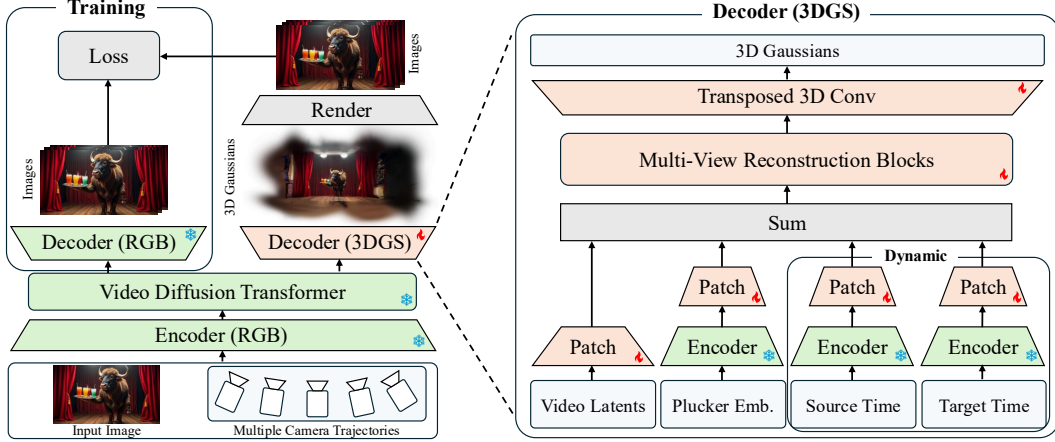


Figure 4: **3D Generative Reconstruction Framework.** Our pipeline builds upon a camera-controlled video diffusion model (Ren et al., 2025) pre-trained on large scale data. We train a 3D Gaussian Splatting (3DGS) decoder by aligning the 2D image renderings of generated 3DGS scenes with the RGB-decoded generations of the pre-trained video model. We only train the 3DGS decoder while freezing the pre-trained autoencoder and diffusion model. At inference time we directly use the 3DGS decoder, without requiring the RGB decoder anymore. Time conditioning within the 3DGS decoder allows us to easily extend our approach from static to dynamic 3D scene generation.

i.e. $\mathbf{z} = \mathcal{E}(\mathbf{I}) \in \mathbb{R}^{L' \times C \times h \times w}$. The training and inference of the diffusion model are then performed in this latent space. The final video $\hat{\mathbf{I}} = \mathcal{D}_{rgb}(\mathbf{z})$ is decoded with a pre-trained VAE decoder $\mathcal{D}_{rgb}$. In this paper, we adopt the pre-trained GEN3C (Ren et al., 2025; Cosmos, 2025) model. Specifically, the latent channel dimension is $C = 16$, the temporal dimension is $L' = (L - 1)/\tau + 1$, and the spatial dimensions are $h = H/\sigma$ and $w = W/\sigma$, where the temporal compression factor is $\tau = 8$ and the spatial compression factor is $\sigma = 8$.

3.2 SELF-DISTILLATION

We train the 3DGS decoder $\mathcal{D}_s$ as a student under a teacher–student paradigm, where a camera-controlled video diffusion model $\mathcal{V}$ serves as the teacher. For diverse supervision, we curate a large-scale set of diverse text prompts with large language models (OpenAI, 2025; Bai et al., 2023), generate images I with an image diffusion model (Black Forest Labs, 2024), and expand into multi-view sequences using GEN3C (Ren et al., 2025).

Teacher–student setup. Given an input image I and a sampled camera trajectory $\{\mathbf{C}^t\}_{t=1}^L$, the video diffusion model $\mathcal{V}$ generates a denoised video latent $\mathbf{z} = \mathcal{V}(I, \{\mathbf{C}^t\}_{t=1}^L)$. The latent $\mathbf{z}$ is decoded along two branches: the pre-trained RGB decoder $\mathcal{D}_{rgb}$ produces video frames $\mathbf{I}_{\mathcal{D}_{rgb}} = \mathcal{D}_{rgb}(\mathbf{z})$, while the 3DGS decoder $\mathcal{D}_s$ outputs explicit 3D Gaussians $\mathbf{G}$. Rendered views from 3DGS (*student*) are defined as $\mathbf{I}_{\mathcal{D}_s} = \text{Render}(\mathbf{G}, \{\mathbf{C}^t\}_{t=1}^L)$ and are supervised to match RGB frames $\mathbf{I}_{\mathcal{D}_{rgb}}$ (*teacher*), forming the self-distillation loop.

Multi-trajectory supervision. In our paper, to enlarge viewpoint coverage for the input image, we sample $V = 6$ camera trajectories per input image, as shown in Fig. 3. For each trajectory v , we construct a spatiotemporal cache $\{\mathbf{P}^{t,v}\}_{t=1}^L$ along with $L = 121$ camera poses $\{\mathbf{C}^{t,v}\}_{t=1}^L$. Passing these caches through video diffusion model $\mathcal{V}$ yields latents $\mathbf{z}^v$, which the teacher decodes into RGB frames $\mathbf{I}_{\mathcal{D}_{rgb}}^v = \mathcal{D}_{rgb}(\mathbf{z}^v)$. The 3DGS decoder $\mathcal{D}_s$ learns to fuse these multiple $\mathbf{z}^v$ into coherent Gaussians $\mathbf{G}$, while filling disoccluded regions. Through this self-distillation design, the 3DGS decoder $\mathcal{D}_s$ is trained entirely from synthetic supervision provided by $\mathcal{V}$. Operating in latent space enables efficient aggregation of multiple trajectories, while the explicit Gaussian output $\mathbf{G}$ enforces geometric consistency and supports downstream simulation and real-time rendering.

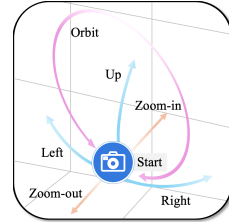


Figure 3: Sampled six camera trajectories to maximize view coverage.

4 FEED-FORWARD RECONSTRUCTION FROM MULTI-VIEW VIDEO LATENTS

Our feed-forward 3DGS decoder $\mathcal{D}_s$ is designed to transform the synthesized multi-view latents generated by our video model $\mathcal{V}$ into an explicit 3D representation that can be rendered from arbitrary viewpoints. We choose 3D Gaussian Splatting (3DGS) (Kerbl et al., 2023) due to its explicit representation and fast rendering speed. We visualize our detailed 3DGS decoder in Fig. 4 (right).

4.1 3DGS DECODER

Scaling Latent-Based 3D Reconstruction. Previous feed-forward reconstruction frameworks generate 3D Gaussians for each pixel, and thus, are limited by the number and resolution of input views they can handle. The contrast in scale is stark: GS-LRM (Zhang et al., 2024e) operates on only 2–4 images at 512×512 resolution, and AnySplat (Jiang et al., 2025a) on 24 images at 448×448 resolution. Our video model, by comparison, synthesizes $V \times L = 6 \times 121 = 726$ input views at the substantially higher 704×1280 resolution—roughly $30\times$ more views at higher resolution than these pixel-space methods, a regime in which they exceed the GPU memory limit outright. The bottleneck lies in the attention mechanism applied to visual tokens, whose memory and compute requirements grow with the number of pixels. To overcome this limitation, we avoid scaling in pixel space and instead operate directly in the compressed video latent space produced by our camera-controlled video diffusion model. The multi-view video latents are denoted as $\mathbf{Z} \in \mathbb{R}^{V \times L' \times C \times h \times w}$.

Architecture. The 3DGS decoder $\mathcal{D}_s$ first maps its inputs into the hidden dimension of the main reconstruction blocks and outputs per-pixel 3D Gaussian features $\mathbf{G} \in \mathbb{R}^{V \times L \times H \times W \times 14}$. Specifically, the inputs are the multi-view video latents $\mathbf{Z}$ and encoded Plücker embeddings $\mathbf{E}$, both with latent dimensions $\mathbf{Z}, \mathbf{E} \in \mathbb{R}^{V \times L' \times C \times h \times w}$. Each input component is first patchified to match the hidden dimension of the main reconstruction blocks. We do not require any additional visual encoder—only a spatial 2×2 patchification layer (Dosovitskiy et al., 2021) to transform the video latents into flattened tokens for the reconstruction network. The sum of both inputs is processed through the reconstruction blocks. We follow the design introduced in Long-LRM (Ziwen et al., 2024), i.e., one block consists of one Transformer (Vaswani et al., 2017) layer followed by seven Mamba-2 (Dao & Gu, 2024) layers. We repeat the block twice to get 16 layers with 512 hidden dimensions. Finally, a transposed 3D convolution maps the hidden representation to 14 Gaussian channels: 3D position (x, y, z) , scale (s_x, s_y, s_z) , rotation quaternion (q_w, q_x, q_y, q_z) , opacity α , and RGB (r, g, b) . Formally, the static decoder is expressed as $\mathbf{G} = \mathcal{D}_s(\mathbf{Z}, \mathbf{E})$.

Plücker embeddings. Raw Plücker embeddings are first computed from camera poses $\{\mathbf{C}^{t,v}\}$ and intrinsics as $\mathbf{E}^{\text{raw}} \in \mathbb{R}^{V \times L \times 6 \times H \times W}$. We reuse the RGB encoder $\mathcal{E}$ from the pre-trained VAE to encode Plücker embeddings. Specifically, we separately encode the 3-dimensional ray directions and the 3-dimensional cross product of ray directions and origins using $\mathcal{E}$, and concatenate latent along the channel dimension to obtain $\mathbf{E}^{\text{enc}} \in \mathbb{R}^{V \times L' \times 2C \times h \times w}$. A lightweight MLP maps $\mathbf{E}^{\text{enc}}$ to $\mathbf{E} \in \mathbb{R}^{V \times L' \times C \times h \times w}$, reducing the channel dimension by a factor of 2 to match video latent $\mathbf{Z}$.

4.2 LOSS FUNCTION

Image-based supervision. We supervise our 3DGS decoder with an image-based reconstruction loss. The reconstruction loss is split into a Mean Squared Error (MSE) loss $\mathcal{L}_{mse}$ and an LPIPS loss $\mathcal{L}_{lpips}$ (Zhang et al., 2018) using VGG (Simonyan & Zisserman, 2014) as a feature extractor.

Depth supervision. We observed that the decoder trained with only RGB loss often produces flattened geometry. To address this, we additionally supervise the rendered depth maps using the consistent video depth estimated by an off-the-shelf system ViPE (Huang et al., 2025), and the scale-invariant depth loss $\mathcal{L}_{depth}$ from Long-LRM (Ziwen et al., 2024).

Opacity-based pruning. Similar to Long-LRM (Ziwen et al., 2024), we use an L1 regularization on the opacity $\mathcal{L}_{opacity}$ and remove the Gaussians with the lowest 80% opacity.

Total loss. Our total loss is computed as the weighted sum of all losses with weight factors λ_i

$$\mathcal{L} = \lambda_{mse} \mathcal{L}_{mse} + \lambda_{lpips} \mathcal{L}_{lpips} + \lambda_{depth} \mathcal{L}_{depth} + \lambda_{opacity} \mathcal{L}_{opacity} \quad (1)$$

where we set $\lambda_{mse} = 1.0$, $\lambda_{lpips} = 0.5$, $\lambda_{depth} = 0.05$, and $\lambda_{opacity} = 0.1$.

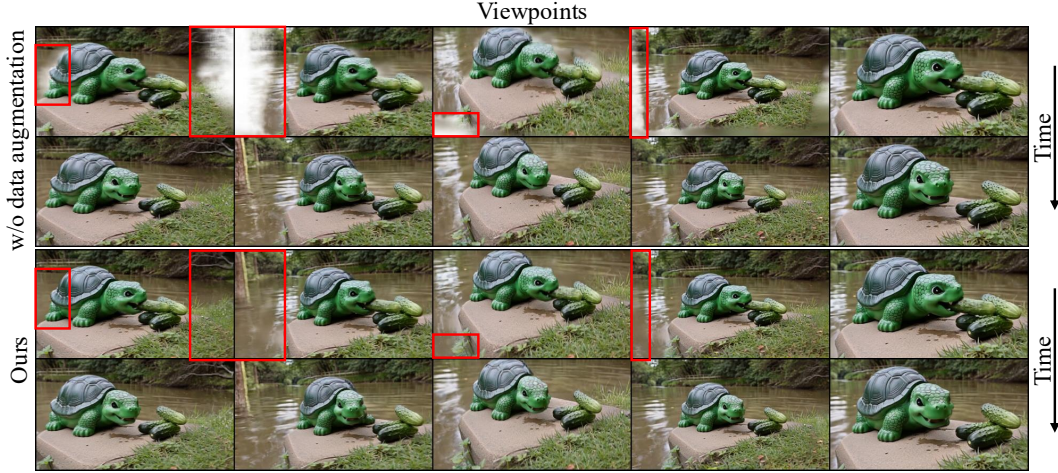


Figure 5: **Dynamic data augmentation.** When naively supervising the time-aware 3DGS decoder, we observe artifacts in the generated 3D Gaussians. Specifically, early timesteps from extreme poses exhibit low opacity in regions not covered by the supervision signal. To address this, we augment the supervision data with a motion-reversed video, ensuring that each timestep is observed from the full spatial coverage, thereby preventing low opacity artifacts in the early timesteps.

5 EXTENSION TO DYNAMIC 3D SCENES

Our approach can be extended to handle dynamic 3D scenes with minimal changes, outlined below.

Self-distillation for dynamic scenes. Our dynamic 3D setup closely follows the design of the static 3D counterpart. Instead of a single image, the video model takes a single video of length L with corresponding camera poses as input and generates multi-view video latents capturing the same underlying motion. For video inputs, we follow a protocol similar to the static case: sampling diverse text prompts with large language models, then generating videos using video diffusion models (Cosmos, 2025; Wan, 2025). The generated videos are then annotated with camera poses and depth maps using ViPE (Huang et al., 2025).

Dynamic 3DGS decoder architecture. We follow the bullet-time design of Liang et al. (2025b) and design our dynamic 3DGS decoder $\mathcal{D}_s$ to generate time-dependent 3D Gaussians for all the video frames. Specifically, the dynamic 3DGS decoder $\mathcal{D}_s$ closely follows the architecture of $\mathcal{D}_s$, except that we augment the $\mathcal{D}_s$ input with the encoded source and target time embeddings $\mathbf{T}^{\text{src}}, \mathbf{T}^{\text{tgt}} \in \mathbb{R}^{V \times L' \times C \times h \times w}$. The unencoded (raw) source times are assigned to each input frame, forming $\mathbf{T}^{\text{src, raw}} \in \mathbb{R}^{V \times L \times 1 \times H \times W}$, and the target time is $\mathbf{T}^{\text{tgt, raw}} \in \mathbb{R}^{V \times 1 \times 1 \times H \times W}$. Both are spatially repeated and augmented with a 2-dimensional sinusoidal embedding to expand the channel dimension to 3, then encoded with the RGB encoder $\mathcal{E}$ to latent dimensions $\mathbf{T}^{\text{src}}, \mathbf{T}^{\text{tgt}} \in \mathbb{R}^{V \times L' \times C \times h \times w}$. The target time is repeated along the latent temporal dimension to match L' . During training, a target time step is randomly sampled, and the corresponding 3D Gaussians are decoded and supervised across all trajectories generated with dynamic data augmentation. We fine-tune $\mathcal{D}_d$ from a pre-trained $\mathcal{D}_s$ with the patchification layers of $\mathbf{T}^{\text{src}}$ and $\mathbf{T}^{\text{tgt}}$ initialized to zeros. $\mathbf{T}^{\text{src}}$ and $\mathbf{T}^{\text{tgt}}$ are directly added to the sum of $\mathbf{Z}$ and $\mathbf{E}$. Formally, the dynamic decoder is defined as $\mathbf{G} = \mathcal{D}_d(\mathbf{Z}, \mathbf{E}, \mathbf{T}^{\text{src}}, \mathbf{T}^{\text{tgt}})$.

Dynamic data augmentation. In the static 3D scenes, where time is not a factor, frames from different timesteps can all be used to supervise the 3DGS renderings. In contrast, for dynamic 3D scenes, the 3DGS changes over time, so only frames from the corresponding timestamp are valid for supervision. This restriction can lead to a trivial solution, where the 3DGS at a given timestep simply ignores information from frames at other timesteps (see Fig. 5).

To mitigate this issue, we introduce a dynamic data augmentation strategy that balances supervision across timesteps. Early frames are naturally associated with viewpoints close to the input image, whereas later frames correspond to more distant viewpoints. To counter this imbalance, we augment the training data with paired supervision views for each timestep — one near and one far. Concretely, we reverse the input video in frame order and feed it into the video model $\mathcal{V}$, which produces six additional multi-view sequences. After reversing these sequences back to the original motion order,

Table 1: **State-of-the-art comparisons for single-image-to-3D generation.** Across RealEstate10K, DL3DV, and Tanks-and-Temples, our method (bottom row) attains the best PSNR $\uparrow$, SSIM $\uparrow$, and LPIPS $\downarrow$ in every column, with the largest PSNR gains—3.45 dB on DL3DV and 3.34 dB on Tanks-and-Temples—on the harder out-of-domain benchmarks. Best results per column are in **bold**; “-” marks metrics not reported by the corresponding baseline.

Method	RealEstate10K			DL3DV			Tanks-and-Temples		
	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$
ZeroNVS	13.01	0.378	0.448	13.35	0.339	0.465	12.94	0.325	0.470
ViewCrafter	16.84	0.514	0.341	15.53	0.525	0.352	14.93	0.483	0.384
Wonderland	17.15	0.550	0.292	16.64	0.574	0.325	15.90	0.510	0.344
Bolt3D	21.54	0.747	0.234	-	-	-	-	-	-
Ours	21.79	0.752	0.219	20.09	0.583	0.313	19.24	0.570	0.336

we obtain six trajectories that progress inward (from far viewpoints toward the input). Combined with the original six outward trajectories, this yields 12 supervision views per timestep. Importantly, this augmentation is applied only during training: the reversed trajectories act purely as extra supervision signals to prevent collapse from pruning artifacts and are not required at inference time.

6 EXPERIMENTS

6.1 EXPERIMENTAL SETUP

Datasets. To train our model, we use no existing multi-view datasets; instead, we construct our own—the *Lyra* dataset—relying solely on our video model to supervise the 3DGS decoder. The scale is substantial: the 3D setup builds on 59,031 images and the 4D setup on 7,378 videos, all sourced from diverse text prompts spanning indoor and outdoor environments, humans, animals, and both realistic and imaginative content. Synthesizing six camera trajectories per image (3D) or video (4D) expands this into 354,186 training videos for 3D and 44,268 for 4D—a $6\times$ multiplication of viewpoint coverage obtained without any real-world capture.

Baselines. We benchmark against four state-of-the-art generative single-image-to-3D methods spanning the field’s main design paradigms: ZeroNVS (Sargent et al., 2024), ViewCrafter (Yu et al., 2024b), Wonderland (Liang et al., 2025a), and Bolt3D (Szymanowicz et al., 2025b), with Wonderland and Bolt3D being the strongest of the four on the benchmarks where they report results. Because *no source code is available* to re-run these methods on our out-of-distribution evaluation set, we compare against their published quantitative results under a matched evaluation protocol, ensuring a like-for-like comparison on the same benchmarks and metrics.

Evaluation. Following prior work, we evaluate single-image-to-3D on three benchmarks of increasing difficulty—RealEstate10K (Zhou et al., 2018), DL3DV (Ling et al., 2024b), and Tanks-and-Temples (Knapitsch et al., 2017)—under the exact evaluation protocol of Wonderland (Liang et al., 2025a) and Bolt3D (Szymanowicz et al., 2025b), so that our numbers are directly comparable to theirs. Performance is reported with the three standard reconstruction metrics used across this literature: PSNR and SSIM (higher is better) and LPIPS (lower is better).

6.2 MAIN RESULTS

Quantitative results. Tab. 1 reports single-image-to-3D results across three benchmarks and three metrics—nine comparison cells in total—and *Lyra* holds the top position in every one of them. The margins are consistent rather than incidental: on RealEstate10K we improve over the best prior method (Bolt3D) on all three metrics (21.79 vs. 21.54 PSNR, 0.752 vs. 0.747 SSIM, 0.219 vs. 0.234 LPIPS), and the gap widens sharply on the more challenging out-of-domain benchmarks, where we surpass the strongest available baseline (Wonderland) by 3.45 PSNR on DL3DV (20.09 vs. 16.64) and 3.34 PSNR on Tanks-and-Temples (19.24 vs. 15.90), with corresponding gains in SSIM and LPIPS. Relative to the weakest baseline, ZeroNVS, the PSNR improvement exceeds 6 dB on all three datasets. Because these improvements are inherited directly from the video teacher, the clearest path to further gains is a stronger video generative model, from which our reconstructions will benefit automatically. We provide additional quantitative comparisons on the *Lyra* dataset in Appendix C.



Figure 6: **Image-to-3DGS Generation.** Across five rendered views per generated 3DGS scene, our method delivers consistently sharp, view-stable reconstructions, visually reflecting the strong quantitative gains reported in our comparisons.



Figure 7: **Ablations.** We visualize ablation results by rendering the same extreme novel viewpoint after image-to-3DGS generation; depth visualizations and ablations are provided in Appendix C.4.

Qualitative results. The quantitative advantage in Tab. 1 is mirrored visually: we show novel-view renderings from our generated 3DGS scenes in Fig. 1 and Fig. 6 (video results will be released upon acceptance). Consistent with the metric gains, our method produces high-quality novel-view content for 3D/4D scenes in unseen regions while remaining faithful to the input image/video. Side-by-side qualitative comparisons on the *Lyra* dataset are provided in Appendix C.

6.3 ABLATIONS

In Tab. 2, we isolate the contribution of each design choice by removing one component at a time and re-measuring PSNR, SSIM, and LPIPS on out-of-distribution diverse prompts, using our full model (24.77 PSNR / 0.837 SSIM / 0.224 LPIPS) as the reference. Each row therefore reports the cost of dropping a single ingredient. As reference signal we compare 3DGS renderings against the videos generated by our camera-controlled video diffusion model. The two components with the largest impact are multi-view fusion and self-distillation itself, each responsible for a multi-point PSNR swing, as detailed below.

Real data only. Replacing self-distillation with training on only real multi-view datasets—RealEstate10K (Zhou et al., 2018) and DL3DV (Ling et al., 2024b), the corpora used by most prior work—collapses performance to 19.08 PSNR / 0.659 SSIM / 0.413 LPIPS, a 5.69 PSNR drop from our full model. This is by far the single largest degradation among the data and loss ablations, quantifying how limited these datasets are for out-of-distribution scenes and how much self-distillation buys us.

Self-distillation + real data. Adding real data on top of self-distillation yields 24.74 PSNR / 0.823 SSIM / 0.236 LPIPS—statistically on par with, and marginally below, self-distillation alone (24.77 / 0.837 / 0.224). The near-identical numbers are informative: the extra real-world supervision contributes nothing measurable, confirming that self-distillation alone is already di-

Table 2: **Ablation study on the *Lyra* dataset.** Starting from our full model (top, 24.77 PSNR / 0.837 SSIM / 0.224 LPIPS), each row removes a single component. Removing multi-view fusion (17.73 PSNR, -7.04) and self-distillation (real data only, 19.08 PSNR, -5.69) causes the largest drops, while a latent-space decoder is required at all (pixel space runs out of memory, OOM).

g Method	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$
Ours	24.77	0.837	0.224
Data	19.08	0.659	0.413
Self-distillation	24.74	0.823	0.236
Multi-view fusion	17.73	0.704	0.313
Self-distillation + Multi-view fusion	24.74	0.823	0.236
Latent-space decoder	OOM	OOM	OOM

verse and consistent enough to train the reconstruction model.

No depth loss. Removing the depth loss lowers all three metrics (24.31 PSNR / 0.811 SSIM / 0.247 LPIPS, a 0.46 PSNR drop) in addition to producing flat geometry, so the loss both regularizes geometry and slightly improves image-based quality. We visualize depth maps in Appendix C.4.

No opacity pruning. Keeping all Gaussians instead of pruning by opacity slightly lowers quality (24.55 vs. 24.77 PSNR) while producing a far less compact representation. Pruning is therefore effectively free quality-wise and pays off at render time: at $H = 704$, $W = 1280$ a frame renders in 18 ms with pruning vs. 30 ms without, a $1.67\times$ speed-up.

No LPIPS loss. Dropping the LPIPS loss produces the second-largest quality regression among the loss ablations—23.74 PSNR / 0.766 SSIM / 0.370 LPIPS, i.e., a 1.03 PSNR drop and a striking 0.146 increase in LPIPS—showing that this loss is what drives robustness to input inconsistencies and the preservation of high-frequency detail.

No multi-view fusion. Generating 3D Gaussians independently per camera trajectory and naively fusing them into one point cloud is the most damaging architectural choice we tested: performance falls to 17.73 PSNR / 0.632 SSIM / 0.446 LPIPS, a 7.04 PSNR drop, the largest in the entire table. Learning the fusion across trajectories with our reconstruction blocks, where each token attends to the others, is thus essential rather than optional.

No Mamba-2. Replacing our joint Transformer/Mamba-2 blocks with Transformer-only blocks slightly lowers quality (24.58 vs. 24.77 PSNR) while being dramatically slower: a single forward pass with $V = 6$, $L = 121$, $H = 704$, $W = 1280$ takes 3213 ms with joint blocks versus 20922 ms with Transformer-only blocks—a $6.5\times$ speed-up at essentially no quality cost.

No latent-based 3DGS. Operating the 3DGS decoder in pixel space instead of latent space simply does not fit in memory (OOM in Tab. 2). Pixel-space methods such as BTimer (Liang et al., 2025b) are capped at 12 input frames, whereas our latent-space design ingests up to 726 input frames—a roughly $60\times$ larger input—which is precisely what makes the high-view-count regime tractable.

7 CONCLUSION

In this work, we propose *Lyra*, a novel 3D and 4D generation framework relying only on a single image or video input. Instead of collecting multi-view datasets, we introduce camera-controlled video diffusion models as teachers within a self-distillation framework for a student 3DGS decoder. Our 3DGS decoder operates in the latent space of the video model and directly reconstructs 3D Gaussians without any post-processing or optimization. The payoff is measurable across the board: *Lyra* ranks first on all three single-image-to-3D benchmarks and all three metrics, improving PSNR by up to 3.45 dB over the strongest prior baseline, and it extends the same advantage to 4D, outperforming a strong pixel-space baseline by 8.60 PSNR (static) and 2.78 PSNR (dynamic) on our diverse dataset. These consistent margins confirm that the design generalizes across diverse scenes, while the dynamic extension demonstrates the feasibility of 4D generation from monocular video. Currently, the scale and consistency of our generated scenes are bounded by the capacity of our camera-controlled video diffusion model. Hence, further improving the video diffusion model will enable large-scale, consistent scene synthesis. Moreover, investigating the adaptation of auto-regressive techniques (Chen et al., 2024a) into our framework for large-scale generation is an interesting direction for future work. Lastly, modelling motion and tracking information within the reconstruction network, as in concurrent work (Lin et al., 2025), could improve visual motion quality.

8 ETHICS STATEMENT

Like all generative AI technologies, there are risks of misuse of our method, including producing misleading or synthetic 3D/4D content. While such risks exist, the main goal of this research is to contribute to the fields of simulation, robotics, and embodied AI by providing scalable and controllable tools for data generation. In this sense, we expect our most significant contributions to be made through the following capabilities and insights.

- **The ability to enable realistic simulation for embodied agents:** We provide controllable and physically consistent camera motion within generated 3D and 4D environments to support training and evaluating agents that must perceive, navigate, and interact with complex scenes.
- **Improved scalability for data engines:** Our method removes the need for multi-view capture and expensive per-scene optimization, and helps produce high-fidelity, customizable scenes at scale, which is particularly suitable for research in robotics, reinforcement learning, and closed-loop simulation.
- **Better technical understanding:** As an academic study, this work contributes a foundational case study of how camera control interacts with spatiotemporal generative models, which will benefit research in computer vision, graphics, and embodied AI.

While we caution against deceptive or wrongful use of this technology—where misleading and/or unverifiable media could be created with realistic generative outputs—we believe that the adoption of safeguards (e.g., provenance tracking, dataset documentation, and good evaluation practices) will be useful in ensuring that generative models advance science and society in positive ways.

9 REPRODUCIBILITY STATEMENT

To fully reproduce our results and accelerate future research in this area, our training and inference code, model weights, and data for both 3D and 4D generation will be released upon acceptance.

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1242 A ADDITIONAL DETAILS OF VIDEO DIFFUSION MODEL

1243 This section provides additional information on the GEN3C (Ren et al., 2025) video diffusion model
1244 backbone introduced in Sec. 3.1 and how we further improve its 3D consistency.

1245 **Conservative mask refinement.** A key limitation of GEN3C’s forward warping is background
1246 leakage, where occluded object parts in the source view (e.g., lion body in Fig. 8) become visible in
1247 novel viewpoints but are not properly indicated in the disocclusion mask. This leads to incomplete
1248 foreground completion by the video diffusion model, as shown in Fig. 8 (second and fourth columns).

1249 To address this, we adopt a conservative strategy: since the geometry of occluded regions in $\mathbf{P}^{t,v}$
1250 is unknown, we assume all such areas are occupied and derive refined disocclusion masks accordingly.
1251 This allows the video model to reason about potentially visible foreground regions that were missed
1252 by standard point-based rendering.

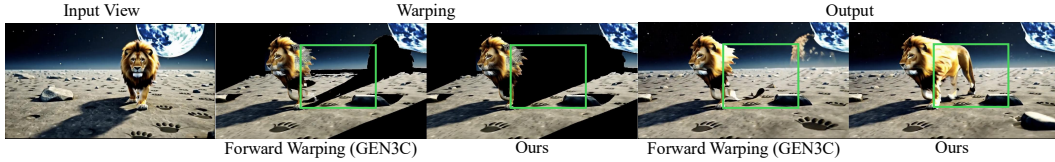
1253 Our approach constructs a triangular mesh from camera-space points $\mathbf{P}_c^{t,v} = \mathbf{C}^t \mathbf{P}^{t,v}$ by connecting
1254 spatially adjacent pixels, similar to Hu et al. (2021):

$$1255 \mathcal{M} = \bigcup_{(u,v)} \{ \triangle(\mathbf{p}_{u,v}, \mathbf{p}_{u+1,v}, \mathbf{p}_{u,v+1}), \triangle(\mathbf{p}_{u+1,v}, \mathbf{p}_{u+1,v+1}, \mathbf{p}_{u,v+1}) \} \quad (2)$$

1256 where $\mathbf{p}_{u,v}$ denotes the 3D point at pixel (u, v) in $\mathbf{P}_c^{t,v}$. This surface mesh serves as a boundary
1257 between visible and invisible regions. For each rendered pixel, we compare the standard point-based
1258 depth $\mathbf{D}^{t,v}$ with the mesh-interpolated depth $\mathbf{D}_{\mathcal{M}}^{t,v}$ obtained via ray-surface intersection:

$$1259 \mathbf{M}^{t,v}(u, v) = \begin{cases} 0 & \text{if } \mathbf{D}_{\mathcal{M}}^{t,v}(u, v) < \mathbf{D}^{t,v}(u, v) - \epsilon \\ \mathbf{M}_{\text{orig}}^{t,v}(u, v) & \text{otherwise} \end{cases}, \quad (3)$$

1260 where $\mathbf{M}_{\text{orig}}^{t,v}$ denotes the original disocclusion mask from forward warping and ϵ is a small tolerance
1261 factor. When the mesh surface is closer than the original points, we conservatively mask out these
1262 regions as potential foreground disocclusions. This produces more reliable guidance for video
1263 generation, especially under large viewpoint changes, as shown in Fig. 8 (third and fifth columns).



1264 Figure 8: **Rendering function comparison.** Our model builds upon GEN3C (Ren et al., 2025) that
1265 uses forward warped images as camera control conditioning. We visualize forward warping (left)
1266 vs. our conservative rendering function (right). The highlighted region is incorrectly filled with
1267 background pixels in forward warping but properly masked in ours, enabling correct completion.

1271 B MODEL DETAILS

1272 **Progressive training.** Directly training the 3DGS decoder on 6 trajectories of 121 frames at
1273 704×1280 is expensive, hence we train our network progressively, as shown in Tab. 3. Furthermore,
1274 we use the final static pre-trained model to initialize our dynamic model. The total training time
1275 takes 6 days with 8 NVIDIA A100 80 GB GPUs. We use gsplat (Ye et al., 2025) as the 3DGS
1276 implementation.

1277 **Subsampling Gaussians.** Generating pixel-aligned 3D Gaussians scales linearly with spatial and
1278 temporal resolution. Our generated scenes span 726 frames at 704×1280 resolution, which at
1279 one Gaussian per pixel would require 654,213,120 Gaussians—far too many to render in real time.
1280 Two compaction steps make this tractable: generating a single 3D Gaussian per 8×8 spatial
1281 neighborhood cuts the count by $64\times$ to 10,222,080, and opacity-based pruning removes a further 80%
1282 to reach 2,044,416—a $320\times$ overall reduction from the naive per-pixel count, yielding a compact yet
1283 high-quality representation.

Table 3: **Progressive training setup.**

Stage	$H \times W$	L	V	S	B	Steps
Static						
1	176×320	17	1	17	4	10k
2	176×320	49	1	49	4	2.5k
3	352×640	49	1	49	2	2.5k
4	704×1280	49	1	49	1	2.5k
5	704×1280	121	1	9	1	57.5k
6	704×1280	121	1–6	9	1	7k
Dynamic						
7	704×1280	121	6	12	1	10k

Joint Transformer-Mamba blocks. We use the reconstruction block design introduced in Long-LRM (Ziwen et al., 2024), i.e., one block of one Transformer layer followed by seven Mamba-2 layers, repeated twice for 16 layers with 512 hidden dimensions. Notably, this is markedly smaller than the networks in Long-LRM (Ziwen et al., 2024) and Wonderland (Liang et al., 2025a), which use 24 layers with 1024 hidden dimensions—i.e., $1.5\times$ the depth and $2\times$ the width—yet, as the main-paper ablation shows, our joint design attains higher quality (24.77 vs. 24.58 PSNR for a Transformer-only variant) at a $6.5\times$ lower inference cost. While pure Transformers are typically more powerful, they are far slower in both training and inference than Mamba-2. Since we trained both configurations under the same budget, we attribute the small remaining gap to the Transformer-only variant needing considerably more iterations to fully catch up in rendering quality.

Camera encoding. We use Plücker embeddings for camera encoding, following prior works (Kant et al., 2024; Zhang et al., 2024e). The embeddings are computed at full spatial and temporal resolution and then projected into the video latent space using the pre-trained RGB encoder. Since the RGB encoder expects 3 input channels but Plücker embeddings are 6-dimensional, we separately encode the 3-dimensional ray directions and the 3-dimensional cross product of ray directions and origins. The resulting latent encodings are then concatenated along the channel dimension.

Time encoding. Similarly, to make the 1-dimensional time input compatible with the 3-channel input expected by the RGB encoder, we concatenate a 2-dimensional sinusoidal embedding to the original time value along the channel dimension. We then replicate these values across the spatial dimensions before encoding the time into the video latent space. This approach allows us to independently encode both the source and target times into the compressed latent representation.

Dynamic data augmentation. We visualize our dynamic data augmentation procedure for two example camera trajectories in Fig. 9. The procedure creates pairs of supervision views that cover the scene from the same motion state but different extreme viewpoints.

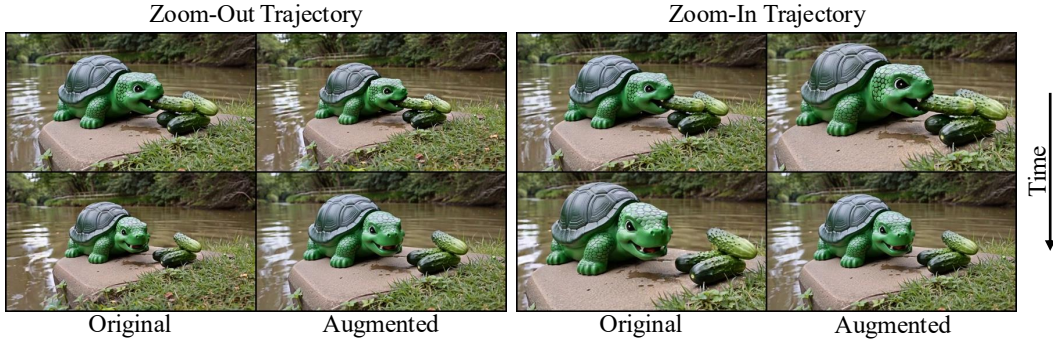


Figure 9: **Dynamic data augmentation videos.** We augment the supervision data with a motion-reversed video, ensuring that each timestep is observed from the full spatial coverage, thereby preventing low opacity artifacts in the early timesteps. We show two example trajectories, i.e., zoom-out and zoom-in, and visualize their corresponding augmented videos. The augmented videos are flipped in their camera motion.



Figure 10: **Image-to-3DGS Comparison.** We compare our method with BTimmer (GEN3C) and visualize five views from generated scenes. We observe significantly fewer artifacts and higher fidelity.

C ADDITIONAL EVALUATION

C.1 ADDITIONAL BASELINE DETAILS

We additionally compare our model against BTimmer (Liang et al., 2025b), a recent approach for joint 3D and 4D reconstruction from posed images or videos. We reached out to the authors of BTimmer to conduct experiments for us using the same scenes as *Lyra*. Since BTimmer is purely regression-based, we integrate it with our GEN3C (Ren et al., 2025) video diffusion backbone. Specifically, we use the same camera trajectories as in our model and provide RGB-decoded videos from GEN3C as input to BTimmer. BTimmer operates in pixel space and requires exactly 12 input images. As discussed in the main paper, pixel-space models such as BTimmer run out of memory on the full 726 high-resolution frames generated by GEN3C—60 $\times$ more frames than they can accept—so we uniformly subsample. For static evaluations, we select 12 frames that evenly cover the available viewpoints; for dynamic evaluations, we sample 11 frames spanning the viewpoint range plus the bullet-time frame corresponding to the motion being reconstructed. This ensures that the target motion state is always present as a reference. We denote this integrated baseline as BTimmer (GEN3C), i.e., BTimmer combined with GEN3C.

C.2 ADDITIONAL STATIC 3D EVALUATION

For image-to-3DGS generation, Fig. 10 shows qualitative comparisons with BTimmer (GEN3C) and Tab. 4a the quantitative ones. The margin is large and uniform across metrics: our method reaches 24.92 PSNR / 0.834 SSIM / 0.183 LPIPS versus 16.32 / 0.580 / 0.427 for BTimmer (GEN3C)—an 8.60 dB PSNR gain and a more than 2 $\times$ reduction in LPIPS—consistent with the visibly higher fidelity and fewer artifacts in the renderings.

C.3 DYNAMIC 3D EVALUATION

BTimmer (GEN3C) comparison. For dynamic 3D evaluation, we compare against BTimmer (GEN3C), as discussed in Sec. C.1, on 100 out-of-distribution videos from our dynamic *Lyra* dataset, cropping to 512×512 for a fair comparison since BTimmer was mainly trained at that resolution. As reported in Tab. 4b, our method leads on every metric—23.07 vs. 20.29 PSNR (+2.78 dB), 0.779 vs. 0.687 SSIM, and 0.231 vs. 0.315 LPIPS—confirming that the advantage seen in the static setting carries over to dynamic scenes.

Table 4: **Quantitative results on the static and dynamic *Lyra* datasets.** On both settings and across all three metrics, our method consistently outperforms the pixel-space BTimer (GEN3C) baseline, with an 8.60 dB PSNR margin in the static case and a 2.78 dB PSNR margin in the dynamic case.

Method	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$
BTimer (GEN3C)	16.32	0.580	0.427
Ours	24.92	0.834	0.183

(a) **Static *Lyra* dataset.** Ours improves PSNR by 8.60 dB (24.92 vs. 16.32) and roughly halves LPIPS (0.183 vs. 0.427) over BTimer (GEN3C).

Method	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$
BTimer (GEN3C)	20.29	0.687	0.315
Ours	23.07	0.779	0.231

(b) **Dynamic *Lyra* dataset.** Ours leads on every metric, improving PSNR by 2.78 dB (23.07 vs. 20.29) and lowering LPIPS by 0.084 (0.231 vs. 0.315) over BTimer (GEN3C).

C.4 DEPTH VISUALIZATION

We visualize depths from our generated 3DGS in Fig. 11. Using depth supervision prevents flat geometries without sacrificing visual quality.



Figure 11: **Depth loss ablation.** We visualize 3DGS renderings and corresponding depth maps. Using the depth loss as additional supervision prevents flat geometries without sacrificing visual quality.

D ROBOT SIMULATION

One of our primary motivations for building a diverse 3D generator is its ability to synthesize simulation environments for autonomous agents. To this end, we develop an end-to-end pipeline: we first generate 3D scenes from text with our model, export the 3D Gaussians as a .ply file, and convert the .ply files into a .usdz format. We use 3DGUT (Wu et al., 2025a) and its export function to create .usdz files. The resulting .usdz file can then be imported into the NVIDIA Isaac robot development platform as a physically based virtual environment. AI-based robots can be trained and tested under diverse conditions within these environments. Early demo results will be released upon acceptance; we include a screenshot of the interface in Fig. 12.

E RELATED WORK

In this section, we will discuss additional related work. We highlight the differences to previous works CAT3D (Gao et al., 2024b) and Bolt3D (Szymanowicz et al., 2025b) in Fig. 13.

3D generation. Early approaches to 3D generation primarily focused on single object categories, extending GANs into the 3D domain by leveraging neural renderers as an inductive bias (DeVries et al., 2021; Chan et al., 2022; Or-El et al., 2022; Schwarz et al., 2022; Bahmani et al., 2023b). With

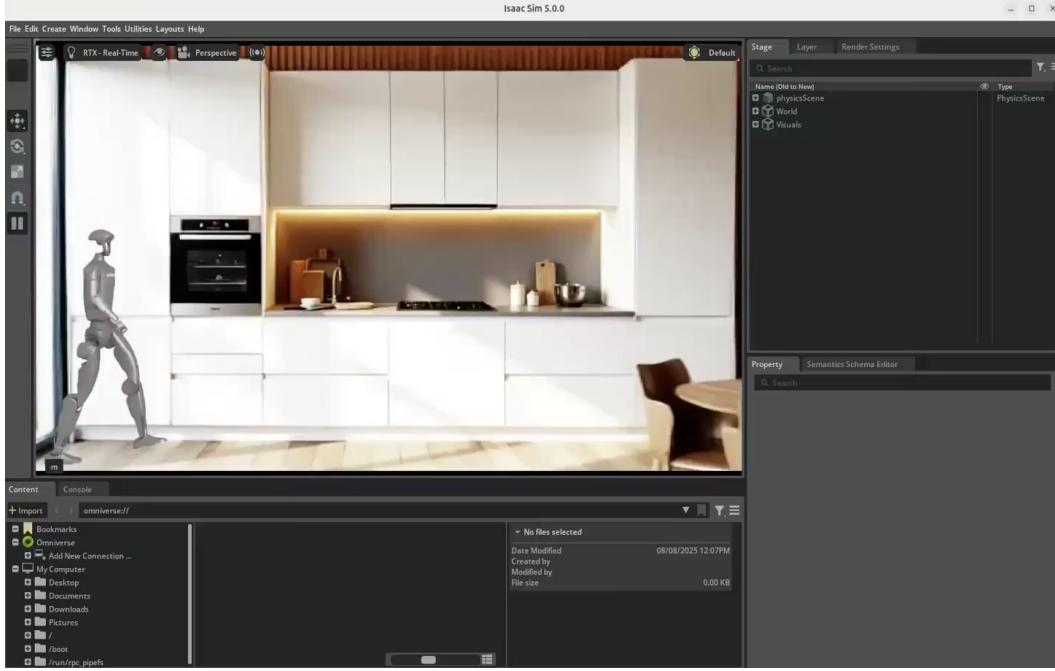


Figure 12: **Robot Simulation.** We visualize a frame of a robot simulation within the Isaac Sim 5.0 simulation framework that takes a generated 3DGS scene from our method as input.

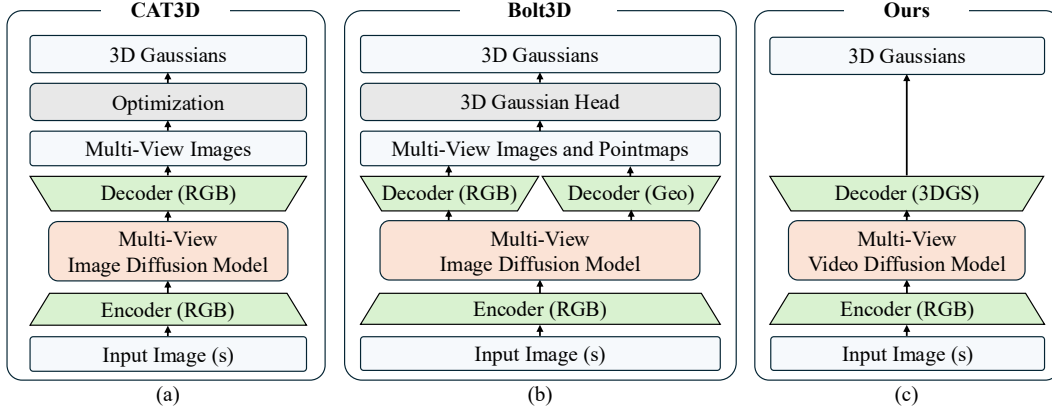


Figure 13: **Approaches for 3D Generation.** CAT3D (Gao et al., 2024b) proposed a multi-view image diffusion model that outputs images from novel viewpoints; subsequently, the images are reconstructed into 3D Gaussians using optimization. Bolt3D (Szymanowicz et al., 2025b) fine-tunes CAT3D to output pointmaps using a geometry autoencoder; instead of relying on optimization, the 3D scene is predicted with a feed-forward 3D Gaussian head. In contrast, our work builds upon a pre-trained camera-controlled video diffusion model and *directly* decodes the multi-view video latents into 3D Gaussians.

the introduction of CLIP-based supervision (Radford et al., 2021), the field advanced toward more flexible and diverse asset creation, enabling both text-driven generation and editing (Chen et al., 2018; Jain et al., 2022; Sanghi et al., 2022; Jetchev, 2021; Gao et al., 2023; Wang et al., 2022). More recently, diffusion models have significantly improved quality by replacing CLIP guidance with Score Distillation Sampling (SDS) (Poole et al., 2023; Wang et al., 2023b; Lin et al., 2023a; Chen et al., 2023b; Liang et al., 2023; Wang et al., 2023a; Li et al., 2024e; He et al., 2024b; Ye et al., 2024; Liu et al., 2024b; Yu et al., 2023; Katzir et al., 2024; Lee et al., 2024a; Sun et al., 2024a).

To further enhance structural coherence, several methods enforce multi-view consistency by generating scenes from different perspectives (Lin et al., 2023b; Liu et al., 2023; Shi et al., 2024; Feng et al.,

2024a; Liu et al., 2024a; Kim et al., 2023; Voleti et al., 2024; Höllein et al., 2024; Tang et al., 2024c; Gao et al., 2024b; Wang et al., 2025f; Kant et al., 2025; Yuan et al., 2025; Ren et al., 2024b), while others adopt iterative inpainting as a strategy for scene synthesis (Höllein et al., 2023; Shriram et al., 2024). Another line of work explores lifting 2D images into 3D using NeRF (Mildenhall et al., 2020), 3D Gaussian Splatting (Kerbl et al., 2023), or mesh-based representations coupled with diffusion models (Chan et al., 2023; Tang et al., 2023a; Gu et al., 2023; Liu et al., 2024c; Yoo et al., 2023; Tewari et al., 2023; Qian et al., 2024b; Long et al., 2024; Wan et al., 2024; Szymanowicz et al., 2023; Lu et al., 2024).

Feed-forward 3D models. Complementary efforts investigate fast, feed-forward techniques that directly predict 3D content from images or text inputs (Hong et al., 2024; Li et al., 2024b; Xu et al., 2024e;d; Zhang et al., 2024a; Han et al., 2024; Jiang & Wang, 2024; Xie et al., 2024a; Tang et al., 2024b; Tochilkin et al., 2024; Qian et al., 2024a; Szymanowicz et al., 2024; 2025a; Liang et al., 2025a; Szymanowicz et al., 2025b; Schwarz et al., 2025; Yang et al., 2025; Zhang et al., 2025a). However, in contrast to our method, these approaches remain limited to producing static 3D scenes. Other methods focus on specific scenes such as faces (Kirschstein et al., 2025). Concurrent work (Liang et al., 2025b; Xu et al., 2025) tackles real scenes but can not handle diverse generated scenes or large viewpoint changes.

4D generation. Recent years have seen rapid progress in 4D generation, i.e., dynamic 3D scene synthesis. Many approaches leverage input text prompts or images as guidance. Following the introduction of large-scale generative models for this task (Singer et al., 2023), subsequent works have made notable advances in both visual fidelity and motion quality (Ren et al., 2023; Ling et al., 2024a; Bahmani et al., 2024b; Zheng et al., 2024b; Gao et al., 2024a; Yang et al., 2024a; Jiang et al., 2024b; Miao et al., 2024; Li et al., 2024a; Zhang et al., 2024g; Yuan et al., 2024; Jiang et al., 2024a).

While text conditioning is widely used, alternative methods aim to lift 2D images or videos into dynamic 3D scenes (Ren et al., 2023; Zhao et al., 2023; Yin et al., 2023; Pan et al., 2024; Zheng et al., 2024b; Ling et al., 2024a; Gao et al., 2024a; Zeng et al., 2024b; Chu et al., 2024; Wu et al., 2024; Yang et al., 2024b; Wang et al., 2024c; Feng et al., 2024b; Sun et al., 2024c; Zhang et al., 2024c; Ren et al., 2024a; Lee et al., 2024b; Li et al., 2024c; Van Hoorick et al., 2024; Uzolas et al., 2024; Chai et al., 2024; Liang et al., 2024; Li et al., 2024d; Xie et al., 2024b; Li et al., 2025a; Wang et al., 2025e; Zhu et al., 2025; Zhou et al., 2025; Hu et al., 2025; Park et al., 2025). Other efforts explore incorporating physics priors into generation pipelines (Lin et al., 2024; Huang et al., 2024b; Zhang et al., 2024f; Kiray et al., 2025), or enhancing motion controllability with template-driven approaches (Zhang et al., 2024d; Sun et al., 2024b; Chen et al., 2024b). A parallel direction emphasizes compositional and interactive 4D generation (Bahmani et al., 2024a; Xu et al., 2024b; Cao et al., 2024; Yu et al., 2024a; Zeng et al., 2024a; Zhu et al., 2024).

Another strand of research extends 3D GAN frameworks into the 4D setting by training directly on 2D video data (Bahmani et al., 2023a; Xu et al., 2023). However, such methods are typically limited by small-scale, single-category datasets, which constrain generalization. Moreover, most existing approaches remain object-centric, often neglecting background elements. As a result, their overall visual fidelity lags behind the high photorealism achieved by recent video generation models, which our method leverages.

Camera-conditioned video models. Recent progress has focused on incorporating camera control into video diffusion models. The pioneering work MotionCtrl (Wang et al., 2024e) introduced camera conditioning by augmenting pre-trained video models (Chen et al., 2023a; Blattmann et al., 2023) with extrinsic matrices. Subsequent efforts (He et al., 2024a; Xu et al., 2024c; Kuang et al., 2024; Ju et al., 2025; He et al., 2025; Li et al., 2025c; Wang et al., 2025c; 2024d; Lei et al., 2025; Liu et al., 2025) enhanced conditioning by representing cameras using Plücker coordinates, while others use follow MotionCtrl to simply use poses flattened (Wang et al., 2025b). Another line of research (Hu et al., 2024; Xiao et al., 2024; Ling et al., 2024c; Hou et al., 2024) enabled camera motion control without introducing additional trainable parameters. Despite these advances, all of the above rely on U-Net-based architectures. Other approaches use low-rank approaches for camera control (Zhang et al., 2025b).

Several recent works continue to extend U-Net-based approaches for camera control (Zhao et al., 2024; Xu et al., 2024a; Zheng et al., 2024a; Zhang et al., 2024b; Yu et al., 2024b; Lei et al., 2024), while others begin exploring diffusion transformers. Cheong et al. (Cheong et al., 2024) propose one such

transformer-based framework, though its scene and visual quality remain limited. DimensionX(Sun et al., 2024d) introduces joint space-time control in diffusion transformers, but relies on pre-defined, non-continuous camera trajectories. Other works (Huang et al., 2024a; Mai et al., 2025; Jiang et al., 2025b) instead explore pose and geometry estimation with a video DiT. CAT4D (Wu et al., 2025b) proposes a multi-view video diffusion model derived from fine-tuning a multi-view backbone. Large-scale synthetic data generation further boosts performance, as demonstrated by SynCamMaster (Bai et al., 2025b) and ReCamMaster (Bai et al., 2025a), which achieve strong results on camera-controlled video generation using transformer-based architectures. Finally, 4DiM (Watson et al., 2024) trains a space-time diffusion model from scratch for novel view synthesis from a single input image. Another line of work edits camera movements of videos with underlying motion using camera retargeting (Seo et al., 2025; Ma et al., 2025). One popular direction has been to use a spatio-temporal cache, e.g., using point clouds or optical flow, to condition the camera-controlled video generation (Wu et al., 2025c; Ren et al., 2025; Wang et al., 2025d; Cao et al., 2025; YU et al., 2025; Jin et al., 2025; Xing et al., 2025; Yan et al., 2025; Li et al., 2025b; Gu et al., 2025).